1.

```
sqlite> create table users(id INTEGER primary key autoincrement,name varchar(45), email varchar(45) unique);
sqlite> create table rooms(id INTEGER primary key autoincrement,name varchar(45), capacity int);
```

```
sqlite> CREATE TABLE reservations(id integer primary key autoincrement, user_id int, room_id int, date date, foreign key(user_id) references users(id), foreign key(room_id) references rooms(id), UNIQUE(ro
om_id, date ));
sqlite> .tables
reservations  rooms         users
```

2.

```
sqlite> insert into users (name, email) values("Raymond", "raymond@ntu.edu.tw");
sqlite> select * from users where name = "Raymond";
1|Raymond|raymond@ntu.edu.tw
```

3.

```
sqlite> insert into users (name, email) values("Mitchell", "mitchell@ntu.edu.tw");
sqlite> insert into users (name, email) values("Annie", "annie@ntu.edu.tw");
sqlite> select * from users where name = "Mitchell" or name = "Annie";
2|Mitchell|mitchell@ntu.edu.tw
3|Annie|annie@ntu.edu.tw
```

4.

```
sqlite> insert into rooms (name, capacity) values("101", 4);
sqlite> insert into rooms (name, capacity) values("102", 6);
sqlite> insert into rooms (name, capacity) values("201", 6);
sqlite> insert into rooms (name, capacity) values("202", 6);
sqlite> insert into rooms (name, capacity) values("301", 12);
```

```
sqlite> select * from rooms;
1|101|4
2|102|6
3|201|6
4|202|6
5|301|12
```

5.

```
sqlite> insert into reservations (user_id, room_id, date) values((select id from users where name = 'Raymond'), (select id from rooms where name=301), '2021-12-12');
sqlite> select * from reservations;
id          user_id     room_id     date
----------  ----------  ----------  ----------
1           1           5           2021-12-12
```

6.

```
sqlite> insert into reservations (user_id, room_id, date) values((select id from users where name = 'Mitchell'), (select id from rooms where name=102), '2021-12-12');
sqlite> select * from reservations;
id          user_id     room_id     date
----------  ----------  ----------  ----------
1           1           5           2021-12-12
2           2           2           2021-12-12
```

7.

```
sqlite> insert into reservations (user_id, room_id, date) values((select id from users where name = 'Raymond'), (select id from rooms where name=101), '2021-12-25');
sqlite> select * from reservations;
id          user_id     room_id     date
----------  ----------  ----------  ----------
1           1           5           2021-12-12
2           2           2           2021-12-12
3           1           1           2021-12-25
```

8.

```
sqlite> insert into reservations (user_id, room_id, date) values((select id from users where name = 'Annie'), (select id from rooms where name=202), '2021-12-25');
sqlite> select * from reservations;
id          user_id     room_id     date
----------  ----------  ----------  ----------
1           1           5           2021-12-12
2           2           2           2021-12-12
3           1           1           2021-12-25
4           3           4           2021-12-25
```

9.

```
sqlite> select date, rooms.name as room_name, users.name as user_name, email from reservations join rooms on reservations.room_id = rooms.id join users on reservations.user_id=users.id where date='2021-12
-12';
date        room_name   user_name   email
----------  ----------  ----------  -----------------
2021-12-12  301         Raymond     raymond@ntu.edu.tw
2021-12-12  102         Mitchell    mitchell@ntu.edu.t
```

10.

```
sqlite> select count(user_id) from reservations where user_id=(select id from users where name='Raymond') and date between '2021-12-01' and '2021-12-31';
count(user_id)
--------------
2
```

11.

```
sqlite> update reservations set room_id = (select id from rooms where name=202) where user_id=(select id from users where name='Raymond') and date='2021-12-25';
Error: UNIQUE constraint failed: reservations.room_id, reservations.date
sqlite>
```

會發生error的原因為room_id以及date為unique, 又因為在同一時段同一教室有被預約, 故Raymond無法更改。

12.

```
sqlite> update rooms set capacity=capacity*2 where id =(select id from rooms where name=101);
sqlite> update rooms set capacity=capacity*2 where id =(select id from rooms where name=102);
sqlite> select * from rooms;
id          name        capacity
----------  ----------  ----------
1           101         8
2           102         12
```

13.

```
sqlite> select * from reservations where user_id=(select id from users where name='Raymond');
id          user_id     room_id     date
----------  ----------  ----------  ----------
1           1           5           2021-12-12
3           1           1           2021-12-25
sqlite> delete from reservations where user_id=(select id from users where name='Raymond') and date='2021-12-25';
sqlite> select * from reservations where user_id=(select id from users where name='Raymond');
id          user_id     room_id     date
----------  ----------  ----------  ----------
1           1           5           2021-12-12
```